

1 Introduction

Each spring, in ecosystems across the temperate zone, ecological communities re-assemble from their period of winter dormancy with the hatching of insects, arrival of migratory birds and emergence of new seedlings, leaves and flowers. The seasonal timing of these life cycle events—their phenology—can be quite different, even among closely related organisms in a shared habitat (Ovaskainen *et al.*, 2013; An *et al.*, 2022). These phenological differences among taxa have often been overlooked in ecological theory outside of plant-pollinator interactions, but growing evidence suggests that they may play a major role in dictating community interactions and shaping the structure and function of ecological communities (Cleland & Wolkovich, 2024; Blackford *et al.*, 2020; Rudolf, 2019). In particular, experiments show that the order in which species begin their seasonal growth can alter the outcomes of competitive interactions (Dawson-Glass *et al.*, 2025; Buonaiuto & Wolkovich, 2023), and related theory suggests that phenology may determine community assembly (Wolkovich & Donahue, 2021; Levine *et al.*, 2022).

Such phenological differences among species can generate seasonal priority effects where earlier active species gain a competitive advantage. This advantage may occur through access to limiting resources before other competitors emerge or arrive, and other ways that modify the environment to the detriment of later emerging species (Fukami, 2015; Weidlich *et al.*, 2020). Studies in plants (Wainwright *et al.*, 2011; Koerner *et al.*, 2008), amphibians (Rudolf & Singh, 2013), insects (Ekholm *et al.*, 2020; Rasmussen *et al.*, 2014) and birds (Samplonius & Both, 2019) have found that even small phenological priority effects, on the order of just a few days, can substantially alter competitive hierarchies in ways that can persist across several growing seasons (Vaughn & Young, 2015).

The demonstrated importance of seasonal priority effects in mediating interspecific competition has led to growing interest in the role of seasonal priority effects in coexistence (Diez *et al.*, 2020; Blackford *et al.*, 2020). Several existing community assembly models that integrate the timing of species arrival or activity suggest timing is fundamental to coexistence (Rudolf, 2019), with multiple studies proposing a fundamental trade-off between early phenology to gain unrestricted access to resources versus later phenology, which requires competing well under low resource conditions (Wolkovich & Donahue, 2021; Levine *et al.*, 2022). Growing interest in understanding seasonal priority effects also highlights one of the major challenges to the field: phenology is not a static trait. Phenology varies highly across years because of the contribution of the environment in determining the timing of event for most, if not all, species. Functionally, phenology is the realized product of an interaction between the environment (cues) and species unique physiology, genetic and developmental attributes (Donohue *et al.*, 2005). With environmental cues changing rapidly in response to anthropogenic climate change, phenological shifts are one of the most pronounced indicators of climate change to date (Scranton & Amarasekare, 2017). However, because the environment is only one part of phenology, species' phenological responses to climate change often differ significantly from each other.

Differences in how species respond to the same environmental cues are one of the primary mediators of the trade-off between phenology and competition and they are likely amplified by climate change. Understanding

the degree to which a species responds to environmental variation, known as **phenological sensitivity** (Δ phenology/ Δ environmental cue), is key to predicting community assembly in an era of global change. Below we outline why integrating phenological sensitivity into community assembly is critical, and outline a path to develop a more robust way to predict assembly in a rapidly changing world.

2 How phenological differences arise

Environmental cues such as temperature, light and moisture are the primary triggers that control phenology (Forrest & Miller-Rushing, 2010). Plants and animals may rely on different cues (Lang *et al.*, 2025; Chuine & Régnière, 2017), and many species respond to multiple aspects of cues. For example, many temperate plants that rely on temperature as a cue require exposure to both cool winter (0-4°C, hereafter: winter chilling) followed by warm spring temperatures to initiate phenological transitions (Ettinger *et al.*, 2020). These sensitivities are the unique product of species physiology, genetics and development that determine phenology depending on the exact environment. Thus, *realized phenology differences* between species will vary substantially depending on the environmental conditions.

Decades of seed germination studies provide reliable examples of how realized phenology varies across species $\times$ environment (e.g., Fernández-Pascual *et al.*, 2021). We illustrate this point by re-analyzing seed phenology data from Buonaiuto (2023), see extended Methods for details), in which 11 species from one habitat were germinated under varying temperature conditions to determine their sensitivities. These co-occurring species showed very different sensitivities to the same environmental cues (Figure 1a) that could potentially impact competition.

For example, for *Persicaria virginiana* and *Eurybia divaricata*—a pair of woodland herbs that frequently compete in Eastern Temperate Forests—which species germinates first depends on environmental conditions, and the order switches with increasing cue levels (Figure 1b,c. in this case, chilling duration). This illustrates that the strength of the seasonal priority effect is highly dependent on the environment, yet this is not how most coexistence models approach phenology.

3 Recent work integrating phenology into assembly models

Growing interest in how phenology may structure communities has led to increased modeling efforts, with most approaches centering on the relationship between timing and competition (Rudolf, 2019; Wolkovich & Donahue, 2021; Levine *et al.*, 2022). Building off experimental (Godoy & Levine, 2014) and observational (Wolkovich, 2013) evidence that earlier species appear to perform better, models have generally built in trade-off where species with a timing advantage also experience greater competition. Models have done this through an explicit connection of phenological differences to parameters controlling the strength of competition (Rudolf, 2019), or implicitly by providing a fitness advantage to earlier species that then must

be offset in communities where species also vary in parameters related to R^* (Wolkovich & Donahue, 2021; Levine *et al.*, 2022).

Models providing a fitness advantage to earlier species have all recognized the importance of environmental variability in communities structured in part by phenological differences, generally including interannual variation in the environment that in turn drives differences in phenology. Many models—focused often on grasslands and water competition—have assigned phenology as equivalent to an intrinsic species-level timing (or season length strategy). Year to year variation in the environment then alters absolute differences between species, which then consistently co-varies with resource competition (Godoy & Levine, 2014; Levine *et al.*, 2022). In these approaches the relative orders between species remain fixed, but other models allow species relative orders and timings to vary year-to-year assuming the underlying relationship between pairwise differences in timings and interspecific-competition is known—and static (Rudolf, 2019). Each of these approaches thus fixes one aspect of what is actually a dynamic environment $\times$ phenology $\times$ competition relationship (i.e., either through an explicit fixed relationship between phenology and competition or fixing the order of species but allowing the strength of competition to vary). This is understandable simplification, but one that may make forecasting community shifts with climate change challenging, as the environment becomes more variable.

Community assembly models including phenology have all included environmental variability (Rudolf, 2019; Levine *et al.*, 2022) assuming an underlying stationary environment (one that varies year-to-year but with an underlying constant distribution), but climate change represents a clearly non-stationary environment. Temperature, for example, trends higher in its mean over time, with potential additional changes to the variance (Canadell *et al.*, 2021). Because environmental variability in assembly models structures what species combinations are stable, shifts in the environment with anthropogenic climate change are likely to reshape communities, but forecasting them requires accurately capturing how the environment affects phenology and competition together in a simple—but still robust—form. We argue this means capturing how dynamically the environment can determine both absolute and relative timings of species, and the how those timings directly impact co-existence. While current modeling approaches to date have simplified one aspect of this problem, we expect that it can be addressed through new models that explicitly allow these dynamics in both inter- and inter-annual time, and connect to the underlying physiology of phenology.

4 Climate Change, Phenology and Coexistence

Leveraging community assembly models to address how shifting phenology with climate change may affect coexistence requires models that encapsulate the strong inter-dependence between environmental variability to competition strength via phenology. While many coexistence models include inter-annual environmental variability, including it intra-annually as well would allow modeling phenology more physiologically, which would better capture how the competition depends on the environment filtered through species phenological

sensitivities. This modeling approach may also show how climate change can disrupt the stability of trait-based, community trade-offs.

We extended a traditional assembly model to include phenological sensitivity in its simplest form—two species with differential sensitivity to one environmental cue. Our model was similar to others in varying species intrinsic competitive abilities at the same time we varied phenology, but differed in modeling phenological sensitivities, making phenology each year a direct response to the intra-annual environmental variation. Comparing long-term co-occurrence (which we consider coexistence here) of species pairs under two different climate regimes we found that each required totally different trait variation such that no pair co-existing under one climate could also co-exist under the other (Figure 2a, see circled points for example, results from 1000 runs of 300-year duration under two different climate normals; see Extended Method for details). This is because the relative phenological differences between species that structures the tradeoff are dependent on the interaction between each species’ phenological sensitivity and the environment itself, and therefore, in a novel environment the relationship between differences in species phenological sensitivities and realized phenological differences shift (Figure 2b). Thus, for any species that currently coexist due to the tradeoff between relative phenology and intrinsic competitive ability, climate change will disrupt this coexistence mechanism.

Our modeling extension revealed an additional, important point for improving the predictive power of coexistence model with climate change. While the tradeoff between realized phenological differences and competitive differences is stable across a shifting environment (i.e., the same magnitude if phenological differences are required to offset the same level of differences in competitive ability), it is the tradeoff between phenological sensitivity differences and competition differences that change in a novel environment (Figure 2a,c). Thus, measuring species phenologies may far less useful for understanding future community assembly than measuring their phenological sensitivities.

Phenological sensitivity is a trait that can be readily measured for real species, providing a link between assembly theory and observed ecological changes. Yet, effectively leveraging this trade-off between phenological sensitivity and competitive ability to model community assembly under future climates will require major efforts by the modeling and phenology-focused communities. Useful predictions will need the continued development of both novel experimental approaches and more-biologically realistic modeling frameworks, but, perhaps most importantly, increased integration between the two.

5 Paths Forward

On the experimental side, more studies that quantify species-level phenological sensitivity could help answer open questions in ecophysiology with major implications for forecasting. Studies that estimate the phenological sensitivity to multiple environmental cues could address how much local adaptation versus plasticity structure phenological sensitivity and competitive traits (e.g. (Lipowsky *et al.*, 2015; Liang & Wu, 2021), but

will be most useful for forecasting if they include multiple environmental cues and a greater scope of taxa and lifeforms. Estimates of species-level phenological sensitivity, quantified at multiple scales (i.e., population, individual) would allow researchers to parameterize community assembly models for species of interest. This could form the start of practical application of these models, building the groundwork for modelers and experimentalists work together to build predictive tools that aid ecological practitioners. But this will require expanding the study systems and flexibility of current models by both research groups.

Models of ecological competition that include phenology are often developed to encapsulate competition between two annual plants (Godoy & Levine, 2014; Wolkovich & Donahue, 2021; Levine *et al.*, 2022). However, herbaceous perennials and woody plants are the dominant functional types in many ecological systems, and, as such, phenology-mediated competition occurs between individuals both within- (i.e., seedling vs. seedling or mature plant vs mature plant) and across- (i.e, mature plant vs. seedling) life stages. While studies in woody plants suggest separate phenological phases can respond quite differently to the same environmental cues (Buonaiuto & Wolkovich, 2021) and that juveniles and adults express substantially different phenological patterns in the same environments (Vitasse, 2013), changes in phenological sensitivity to multiple cues across multiple life stages relevant to community assembly (i.e., seed germination vs vegetative cues) have not been widely explored across taxa. To help, modelers can construct more biologically realistic community assembly models that include multiple growth forms and life stages. At the same time, experimentalists could design studies that explore how environmental variability influences phenological differences across life-stages and mediates both intra- an inter-specific competition within and across these stages.

On the modeling side, researchers should integrate the more complex cue responses to assembly models. Current models—including those conceptualized for seed germination—include phenology independent of or weakly linked to the environment, but research has repeatedly shown than phenology is determined to multiple, interacting environmental cues. For seed germination this often includes multiple aspects of temperature (i.e., forcing and chilling), light (daylength or irradiance) and moisture (Baskin & Baskin, 2014; Forrest & Miller-Rushing, 2010), but models to date have ignored this. Selection for interacting cues is generally thought to be driven by another component likely critical for better predictions—environmental risk.

Complex cues can reduce risk (Bonamour *et al.*, 2019) but they also result in a more conservative phenological strategy. For example, leafout of many forest trees relies on cues of winter chilling and/or photoperiod, which could reduce the risk of frost damage in unstable spring environments, but generally also results a later phenological transitions (Zohner *et al.*, 2017). Ecological models generally subsume such risks into species traits (e.g., increasing maintenance costs), which misses the reality that such risks are also product of timing $\times$ environment.

A changing risk landscape associated with a changing environment may be the underlying driver of observed correlations between the timing and intensity component of life cycle transitions (i.e., germination phenology and total germination percent of a seed cohort Baskin & Baskin, 2014). But models will need to explicitly

include the environment and phenology to predict this. Further, for plants—and likely other organisms—the timing and fraction of phenological success (e.g., seed germination or leafout) usually covary in nature, which likely has important implications for coexistence. Building models that can isolate these two trade-off axes of coexistence—timing and abundance (fraction)—would allow offer to assess the relative importance of these within- and across season trade-offs.

Another dimension of risk and phenology could help generate more predictive assembly models is to understand how another aspect of phenology that is sensitive to environmental cues—the duration of a phenological phases—mitigates or exacerbates environmental risk. Environmental cues not only advance or delay the start of a phenological transition, but may also influence the rate of the transition. More protracted phenological phases (i.e., if the germination of a full seed cohort takes longer) associated with unfavorable conditions may serve as a form of within-season bet-hedging in which an untimely mortality event would have a lower fitness cost to a cohort of seed that started germinating early but slowly (at the cohort level) than one that started early and finished fast. Introducing biologically realistic mortality events into community assembly models that represent important selective forces, such as late-season frosts or early droughts, offers a straightforward approach to assess the importance of the often overlooked aspect of phenology.

The modeling framework we present has the flexibility to incorporate many of the ecological aspects we discussed above (multiple cues, environmental risk, within- and across- season bet-hedging). But making these predictions meaningful will require modelers and physiologists working together to further develop models with biologically accurate physiological processes, and experiments that can readily inform model parameters and development. In a rapidly changing world, these kind of cross-disciplinary collaborations are critical to translate ecological theory into practice and advance a more predictive ecology than is better equipped to inform ecosystem conservation and management.

Figures

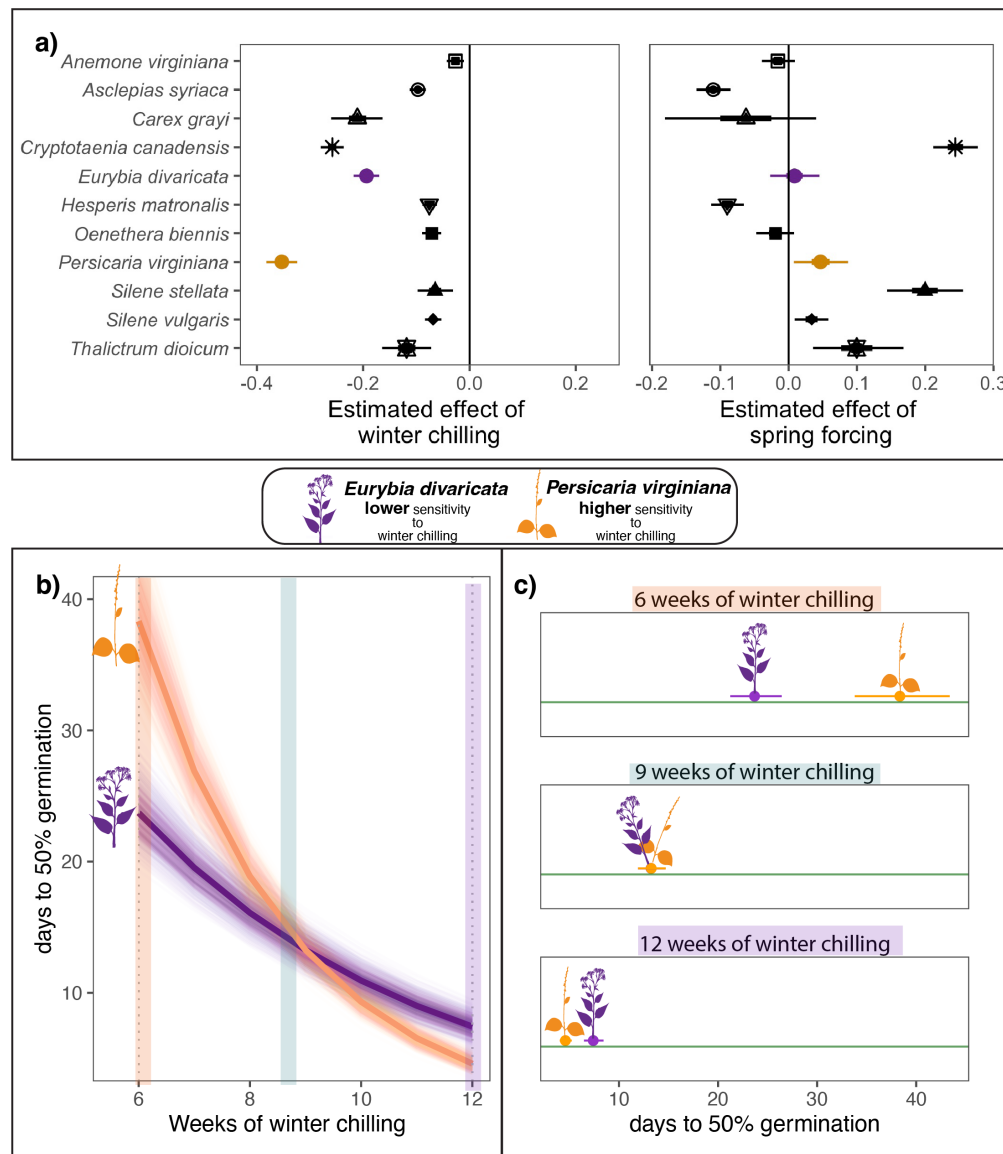


Figure 1: **Co-occurring species can differ substantially in their phenological sensitivity (Δ phenological event/ Δ climate) to the environment.** Panel a) estimates the phenological sensitivity of germination timing to winter chilling and spring forcing for eleven herbaceous species of eastern North America. Points represent the mean estimate and thick and thin bars represent the 50 and 95% uncertainty intervals. In b) and c) we highlight the phenological sensitivity to winter chilling of two species *Eurybia divaricata* and *Persicaria virginiana* that are commonly observed in competition. b) Depicts the sensitivity of each species to winter chilling, with thick lines representing the mean estimate and lighter lines representing modeled uncertainty. Panel c) illustrates how the species' different sensitivities manifest in contrasting realized germination differences under different climate conditions. Points represent the predicted mean number of days to 50% germination and bars the 95% uncertainty intervals

References

- An, S., Chen, X., Shen, M., Zhang, X., Lang, W. & Liu, G. (2022). Increasing interspecific difference of alpine herb phenology on the eastern qinghai-tibet plateau. *Frontiers in Plant Science*, Volume 13 - 2022.

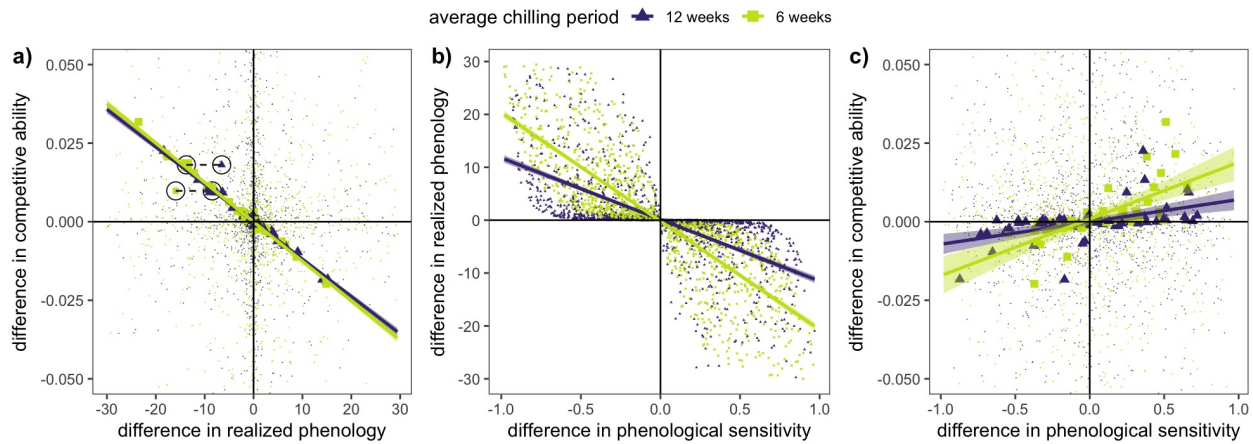


Figure 2: Climate change will alter the relationship competitive ability, phenological sensitivity, realized phenology and coexistence. Panel a) depicts the trade-off between realized phenological differences and competitive differences. While the tradeoff relationship between realized phenology and competition does not shift with climate change, species that coexisted under one climate scenario will not coexist if climate conditions shift, as highlighted by the two sets species pairs circled in panel a). These changes occur because the relationship between phenological sensitivity differences and realized phenology differences, depicted in panel b), are dependent on the environment, and a change in chilling conditions alters this relationship. Panel c) depicts the trade-off between phenological sensitivity differences and competitive differences under two climate scenarios (average of 12 and 6 weeks of chilling). With climate change, the trade-off between phenological sensitivity and competition does shift because lower chilling amplifies the realized phenological differences among species.

Baskin, C. & Baskin, J. (2014). *Seeds: Ecology, Biogeography, and Evolution of Dormancy and Germination*. Elsevier Inc.

Blackford, C., Germain, R. M. & Gilbert, B. (2020). Species differences in phenology shape coexistence. *The American Naturalist*, 195, E168–E180.

Bonamour, S., Chevin, L.-M., Charmantier, A. & Teplitsky, C. (2019). Phenotypic plasticity in response to climate change: the importance of cue variation. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 374, 20180178.

Buonaiuto, D.M., . W. E. (2023). Germination phenology and seedling competition between two forest herb species of the eastern united states in controlled environments. *Knowledge Network for Biocomplexity*, doi:10.5063/F1BZ64HV.

Buonaiuto, D. M. & Wolkovich, E. M. (2021). Differences between flower and leaf phenological responses to environmental variation drive shifts in spring phenological sequences of temperate woody plants. *Journal of Ecology*, 109, 2922–2933.

- Buonaiuto, D. M. & Wolkovich, E. M. (2023). Contrasting responses to climate variability generate seasonal priority effects between native and invasive forest herbs. *Journal of Ecology*, 111, 1711–1721.
- Canadell, J., Monteiro, P., Costa, M., Cotrim da Cunha, L., Cox, P., Eliseev, A., Henson, S., Ishii, M., Jaccard, S., Koven, C., Lohila, A., Patra, P., Piao, S., Rogelj, J., Syampungani, S., Zaehle, S. & Zickfeld, K. (2021). *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, New York, NY.
- Chaine, I. & Régnière, J. (2017). Process-based models of phenology for plants and animals. *Annual Review of Ecology, Evolution, and Systematics*, 48, 159–182.
- Cleland, E. E. & Wolkovich, E. (2024). Effects of phenology on plant community assembly and structure. *Annual Review of Ecology, Evolution, and Systematics*, 55, 471–492.
- Dawson-Glass, E., Schiafo, R., Kuebbing, S. E. & Stuble, K. L. (2025). Warming-induced changes in seasonal priority effects drive shifts in community composition. *Ecology*, 106, e4504.
- Diez, J., Schlauch, J., DuCharme, E., Collings, J. & Erskine, S. (????). Germination phenology alters species coexistence outcomes. *Journal of Ecology*, n/a.
- Donohue, K., DORN, L., Griffith, C., Kim, E., Anna, A., Polisetty, C. R. & Johanna, S. (2005). The evolutionary ecology of seed germination of *Arabidopsis thaliana*: Variable natural selection on germination timing. *Evolution*, 59, 758–770.
- Ekholm, A., Tack, A. J. M., Pulkkinen, P. & Roslin, T. (2020). Host plant phenology, insect outbreaks and herbivore communities – the importance of timing. *Journal of Animal Ecology*, 89, 829–841.
- Ettinger, A. K., Chamberlain, C. J., Morales-Castilla, I., Buonaiuto, D. M., Flynn, D. F. B., Savas, T., Samaha, J. A. & Wolkovich, E. M. (2020). Winter temperatures predominate in spring phenological responses to warming. *Nature Climate Change*, 10, 1137–1142.
- Fernández-Pascual, E., Carta, A., Mondoni, A., Cavieres, L. A., Rosbakh, S., Venn, S., Satyanti, A., Guja, L., Briceño, V. F., Vandellook, F., Mattana, E., Saatkamp, A., Bu, H., Sommerville, K., Poschlod, P., Liu, K., Nicotra, A. & Jiménez-Alfaro, B. (2021). The seed germination spectrum of alpine plants: a global meta-analysis. *New Phytologist*, 229, 3573–3586.
- Forrest, J. & Miller-Rushing, A. J. (2010). Toward a synthetic understanding of the role of phenology in ecology and evolution. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 365, 3101–3112.
- Fukami, T. (2015). Historical contingency in community assembly: Integrating niches, species pools, and priority effects. *Annual Review of Ecology, Evolution, and Systematics*, 46, 1–23.

- Godoy, O. & Levine, J. M. (2014). Phenology effects on invasion success: insights from coupling field experiments to coexistence theory. *Ecology*, 95, 726–736.
- Koerner, C., Stoecklin, J., Reuther-Thiebaud, L. & Pelaez-Riedl, S. (2008). Small differences in arrival time influence composition and productivity of plant communities. *New Phytologist*, 177, 698–705.
- Lang, W., Zhang, Y., Li, X., Meng, F., Liu, Q., Wang, K., Xu, H., Chen, A., Peñuelas, J., Janssens, I. A. & Piao, S. (2025). Phenological divergence between plants and animals under climate change. *Nat Ecol Evol*, 9, 261–272.
- Levine, J. I., Levine, J. M., Gibbs, T. & Pacala, S. W. (2022). Competition for water and species coexistence in phenologically structured annual plant communities. *Ecology Letters*, 25, 1110–1125.
- Liang, L. & Wu, J. (2021). An empirical method to account for climatic adaptation in plant phenology models. *International Journal of Biometeorology*, 65, 1953–1966.
- Lipowsky, A., Roscher, C., Schumacher, J., Michalski, S. G., Gubsch, M., Buchmann, N., Schulze, E.-D. & Schmid, B. (2015). Plasticity of functional traits of forb species in response to biodiversity. *Perspectives in Plant Ecology, Evolution and Systematics*, 17, 66–77.
- Ovaskainen, O., Skorokhodova, S., Yakovleva, M., Sukhov, A., Kutenkov, A., Kutenkova, N., Shcherbakov, A., Meyke, E. & del Mar Delgado, M. (2013). Community-level phenological response to climate change. *Proceedings of the National Academy of Sciences*, 110, 13434–13439.
- Rasmussen, N. L., Van Allen, B. G. & Rudolf, V. H. W. (2014). Linking phenological shifts to species interactions through size-mediated priority effects. *Journal of Animal Ecology*, 83, 1206–1215.
- Rudolf, V. H. W. (2019). The role of seasonal timing and phenological shifts for species coexistence. *Ecology Letters*.
- Rudolf, V. H. W. & Singh, M. (2013). Disentangling climate change effects on species interactions: effects of temperature, phenological shifts, and body size. *Oecologia*, 173, 1043–1052.
- Samplonius, J. M. & Both, C. (2019). Climate change may affect fatal competition between two bird species. *Current Biology*, 29, 327–331.e2.
- Scranton, K. & Amarasekare, P. (2017). Predicting phenological shifts in a changing climate. *Proceedings of the National Academy of Sciences*, 114, 13212–13217.
- Vaughn, K. J. & Young, T. P. (2015). Short-term priority over exotic annuals increases the initial density and longer-term cover of native perennial grasses. *Ecological Applications*, 25, 791–799.
- Vitasse, Y. (2013). Ontogenic changes rather than difference in temperature cause understory trees to leaf out earlier. *New Phytologist*, 198, 149–155.

275 Wainwright, C. E., Wolkovich, E. M. & Cleland, E. E. (2011). Seasonal priority effects: implications for
 276 invasion and restoration in a semi-arid system. *Journal of Applied Ecology*, 49, 234–241.

277 Weidlich, E. W. A., Nelson, C. R., Maron, J. L., Callaway, R. M., Delory, B. M. & Temperton, V. M. (2020).
 278 Priority effects and ecological restoration. *Restoration Ecology*, n/a, e13317.

279 Wolkovich, E. & Donahue, M. J. (2021). How phenological tracking shapes species and communities in
 280 non-stationary environments. *Biological Reviews*, 96, 2810–2827.

281 Wolkovich, E. M. (2013). Flowering phenology of native and exotic species across five sites,
 282 <http://knb.ecoinformatics.org/knb/metacat/wolkovich.41/knb>.

283 Zohner, C. M., Benito, B. M., Fridley, J. D., Svenning, J.-C. & Renner, S. S. (2017). Spring predictability
 284 explains different leaf-out strategies in the woody floras of north america, europe and east asia. *Ecology*
 285 *Letters*, 20, 452–460.